

HACHIOJIMA, Japan

WMO: 476780

Lat: 33.1217N

Lon: 139.7792E

Elev: 153

StdP: 99.51

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	4.9	5.8	-4.0	2.7	6.5	-2.7	3.1	7.2	15.0	12.4	13.4	11.3	5.8	280	0.484

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	4.4	30.0	26.1	29.3	25.8	28.7	25.6	26.9	28.8	26.5	28.3	26.1	27.8	4.2	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.3	22.2	27.9	25.9	21.7	27.6	25.6	21.2	27.3	85.5	28.9	83.7	28.4	82.0	27.9	28.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
13.1	11.5	10.2	2.3	31.5	1.4	0.6	1.3	31.9	0.5	32.2	-0.3	32.6	-1.3	33.0		
			0.0	27.6	1.0	0.5	-0.7	28.0	-1.3	28.3	-1.9	28.6	-2.7	29.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	10.2	10.5	12.6	16.1	19.4	22.0	25.7	26.9	25.0	21.5	17.3	12.8
	DBStd	6.12	2.46	2.69	2.68	2.38	1.76	1.80	1.55	1.14	1.92	2.06	2.43	2.69
	HDD10.0	64	27	23	7	0	0	0	0	0	0	0	0	7
	HDD18.3	953	252	218	177	73	9	1	0	0	1	48	173	
	CDD10.0	3122	33	38	89	182	291	359	487	524	451	358	218	92
	CDD18.3	969	0	0	0	5	42	110	228	266	201	101	16	1
	CDH23.3	5977	0	0	0	1	22	218	1683	2457	1357	232	6	0
	CDH26.7	948	0	0	0	0	0	5	231	508	198	7	0	0
Wind	WSAvg	5.1	5.8	5.7	5.7	5.2	4.6	4.6	4.3	3.9	4.8	5.6	5.1	5.4
Precipitation	PrecAvg	3161	185	192	275	232	253	367	206	193	343	451	273	186
	PrecMax	4607	513	490	626	533	854	734	712	633	820	1053	664	436
	PrecMin	2193	56	34	113	76	97	70	3	29	56	136	84	24
	PrecStd	472	106	95	110	97	115	141	153	124	178	208	124	85
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.3	18.8	19.7	22.5	25.0	27.4	30.3	31.2	30.4	27.5	24.2	21.4
		MCWB	16.2	17.1	16.2	18.5	20.7	24.6	26.3	26.5	25.9	24.1	21.8	19.7
	2%	DB	16.9	17.5	18.8	21.3	23.9	26.0	29.5	30.4	29.5	26.4	22.8	19.4
		MCWB	14.5	15.5	16.3	18.2	20.2	23.8	26.0	26.3	25.6	23.3	19.7	16.8
	5%	DB	15.3	16.3	17.9	20.5	23.1	25.2	28.8	29.6	28.7	25.5	21.9	18.0
		MCWB	12.5	13.8	15.4	17.6	19.7	23.4	25.8	26.0	25.2	22.7	18.9	15.0
	10%	DB	14.0	15.0	17.1	19.7	22.3	24.6	28.0	29.0	27.8	24.7	21.0	16.8
		MCWB	11.0	12.2	14.3	16.9	19.4	23.1	25.4	25.7	24.9	22.1	18.1	14.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	17.9	18.6	20.6	23.0	25.3	27.1	27.7	27.0	25.2	22.7	19.7
		MCDB	17.9	18.4	18.9	21.4	23.9	26.3	29.1	29.6	29.0	26.4	23.7	20.8
	2%	WB	15.0	16.2	17.3	19.6	22.0	24.7	26.6	27.1	26.3	24.4	21.0	17.5
		MCDB	16.5	17.2	18.1	20.4	22.6	25.4	28.4	29.0	28.1	25.6	22.0	18.7
	5%	WB	13.1	14.4	16.1	18.6	21.2	24.1	26.2	26.7	25.9	23.6	19.8	15.8
		MCDB	14.9	15.8	17.5	19.7	22.1	24.8	27.8	28.5	27.6	24.9	21.2	17.3
	10%	WB	11.5	12.7	14.8	17.6	20.4	23.6	25.8	26.3	25.5	22.8	18.6	14.4
		MCDB	13.6	14.5	16.6	19.1	21.5	24.3	27.3	28.1	27.2	24.1	20.3	16.3
Mean Daily Temperature Range	5% DB	MDBR	4.6	5.1	5.5	5.2	4.9	3.8	3.8	4.4	4.5	4.3	4.7	4.6
		MCDBR	6.3	6.5	6.1	5.7	5.6	4.2	4.7	5.0	5.1	5.0	5.5	5.9
		MCWBR	6.1	6.6	5.6	4.3	3.9	2.8	2.6	2.7	2.8	3.5	4.5	5.5
	5% WB	MCDBR	6.1	6.3	5.7	4.6	4.1	3.4	4.2	4.3	4.6	4.4	4.8	5.4
		MCWBR	6.6	7.2	6.4	4.9	3.9	2.5	2.5	2.5	2.8	3.8	5.1	6.0
Clear-Sky Solar Irradiance	taub	0.395	0.439	0.487	0.513	0.502	0.503	0.498	0.473	0.456	0.454	0.420	0.396	
	taud	2.232	2.113	1.991	1.977	2.085	2.184	2.230	2.313	2.334	2.249	2.248	2.232	
	Ebn at Noon	798	798	789	789	800	797	798	814	812	777	765	771	
	Edh at Noon	113	140	171	181	164	148	141	128	120	121	111	107	
All-Sky Solar Radiation	RadAvg	2.25	2.83	3.81	4.64	5.15	4.58	5.81	5.93	4.58	3.22	2.46	2.13	
	RadStd	0.22	0.32	0.28	0.48	0.52	0.55	0.68	0.36	0.36	0.29	0.18	0.19	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (3 neighbors)	N/A	N/A	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page